

Rupsa Bose

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EDUCATION

University of Massachusetts Amherst

B.S. in Computer Science and Mathematics, Commonwealth Honors College

Expected May 2028

Major GPA: 4.0

Coursework: Data Structures and Algorithms, Discrete Mathematics, OOP, Probability & Statistics, Calculus III, Linear Algebra

SKILLS

Languages: Python, Java, C, JavaScript, TypeScript, SQL

Frameworks/Libraries: React.js, Flask, Tailwind CSS, Pandas, NumPy, scikit-learn, Matplotlib, YOLOv8, OpenCV, MediaPipe

Tools: Git, Roboflow, Firebase, Jupyter Notebook, DeepFace, Blender, LabelMe

Databases: MongoDB

EXPERIENCE

Computer Vision Intern – TangoEye

June 2026 – August 2026

- Developed YOLOv8 shoe try-on classifier across 4 retail stores, achieving **89.1%** test top-1 accuracy on **4,035** annotated person crops from **19,000+** CCTV frames.
- Built cash counting detection pipeline with counter zone segmentation and custom YOLO detectors, reaching **90.0%** mAP50 and **99.5%** precision on hand-labeled surveillance data.
- Curated multi-store retail datasets, designed LabelMe annotation workflows, and deployed end-to-end inference pipelines for real-time store analytics.

Undergraduate Assistant – Public Interest Technology (PIT), UMass Amherst

Fall 2025 – Present

- Designed and evaluated LLM-based systems to classify university courses into Public Interest Technology (PIT) categories using prompt engineering and model development.
- Analyzed **30,000+** academic records to identify trends in admissions, enrollment patterns, and academic pathways, building Python-based data pipelines and visualizations (Pandas, NumPy) for institutional reporting.
- Building a dynamic database to track PIT-affiliated students and fellows, enabling structured querying and longitudinal tracking for ongoing research use.
- Supporting research paper development and publication based on this classification work, alongside additional institutional and student data analysis projects.

Computer Vision Intern – Kyrotics

Summer 2025

- Developed YOLOv8 and CNN-based computer vision models for real-time object detection and instance segmentation.
- Improved model performance to **95.4%** mAP50 and **94.4%** F1-score through dataset curation, annotation, and hyperparameter tuning.
- Built end-to-end ML pipelines for preprocessing, validation, training, and deployment using Python and OpenCV.

RESEARCH

Independent Study Researcher – PIT@UMass

Spring 2026 – Present

- Analyzing UMass Amherst registration datasets to identify mathematics and quantitative reasoning pathways among English and History majors.
- Applying computational and statistical analysis methods to study relationships between mathematical thinking and computational thinking (CT).

Undergraduate Research Assistant – DARoS Lab

Spring 2026 – Present

- Researching human-following and obstacle-avoidance systems for legged robots using multimodal perception.
- Developing **SURREAL**-based simulation pipeline using SMPL human meshes and Blender rendering to generate synthetic RGB training data for robotic perception.
- Reviewed and synthesized core methodologies from **20+** embodied AI and robotics papers including **TrackVLA**, **CEAR**, and **MM-Nav**.

PROJECTS

ASL Hand Gesture Recognition System

Dec 2025 – Present

- Built a real-time hand gesture recognition system using Python, OpenCV, CNNs, and MediaPipe with transfer learning via **MobileNetV2**.
- Trained and evaluated the classifier across multiple ASL sign classes, tuning the pipeline for low-latency, real-time inference from live webcam video.

BugME – AI Study Companion (HackMIT)

Sept 2025

- Developed an AI-powered study platform with OCR-driven document parsing, conversational AI workflows, and Google Calendar API automation.
- Built the end-to-end pipeline as part of a small team at HackMIT, integrating document ingestion, LLM-based Q&A, and automated scheduling into a single workflow.

Smart Doorbell System

June 2025

- Engineered a Flask and React-based smart doorbell with DeepFace facial recognition, OpenCV pipelines, and secure visitor logging.
- Implemented a full-stack visitor recognition workflow, from live camera capture through facial recognition to logged, timestamped visitor records in the web dashboard.